V2 Rail Guard + Flex Plate

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Build guide for Bangalore FTC teams | goBILDA FTC StarterBot 2026-27 (BIOBUZZ) | 96mm Hogback, 71mm POLLEN

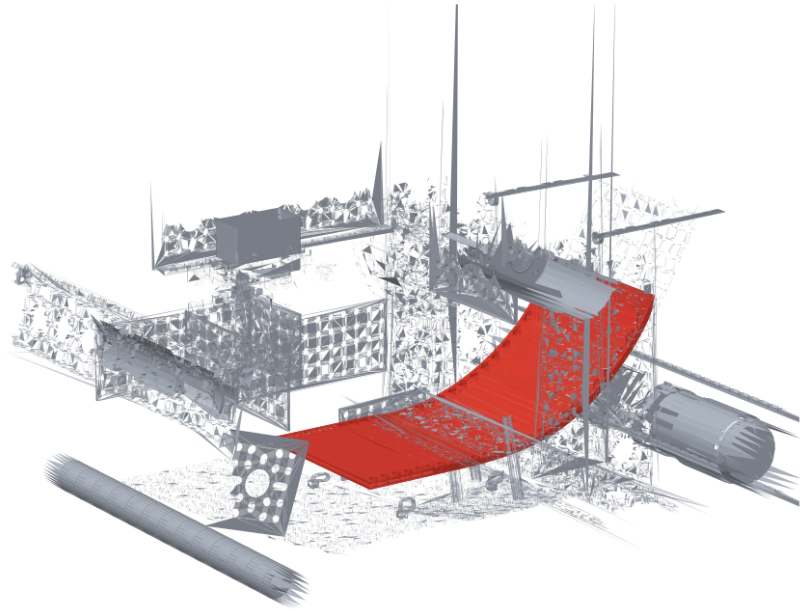
NOT ROBOT TESTED. We have checked this in CAD only. Please do a test print and the hand checks first, and share what you find so we can improve it together.

1. How it works

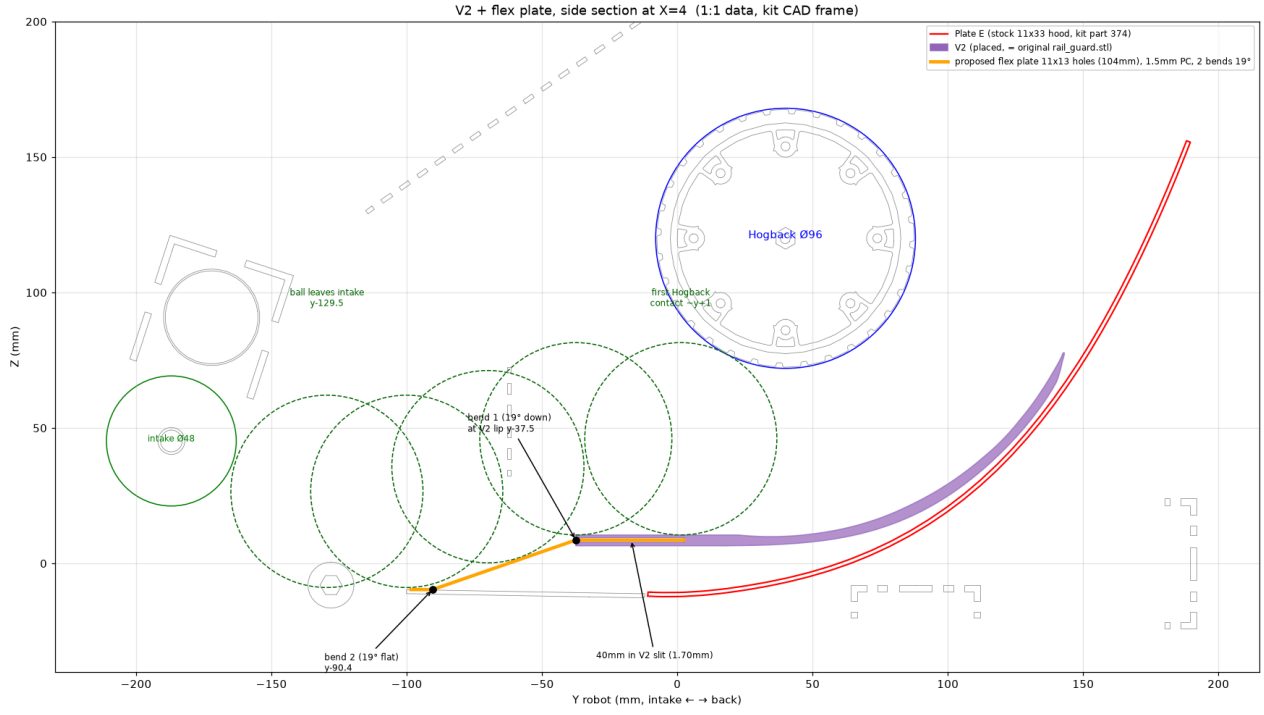
Intake roller ▶ Floor plate ▶ Tail 1 hole ▶ Slope 7 holes ▶ Tongue 5 holes ▶ V2 + Hogback ▶ Shot



Placed in kit CAD (red = V2)



Side section at 1:1 data (orange = flex plate, purple = V2)



2. Key numbers

Item	Value
V2 size	**129.35 x 148.57 x 88mm**, 55 cm3
Squeeze on 71mm POLLEN	**14.5% peak** (12 to 14.5% along curve), target 10 to 15%
Front slot	1.7mm high x 40mm deep (for 1.5mm tongue)
Lip height	20.9mm above floor plate top
Flex plate	11 x 13 holes, 88 x 104mm, two 19 deg bends
Ball climb	18.3mm at 19 deg

3. Parts and tools

Item	Qty	Notes
V2 rail guard (printed)	1	rail_guard_v2.stl
Medium Gridplate 1117-0088-0352	1 per 3 robots	1.5mm polycarbonate, 11 x 44, 8mm grid, 4mm holes, $6.99, goBILDA
Scissors	1	For cutting the plate
Heat gun	optional	Helps the bends
Zip ties	**8**	2 per chain x 4 chains (4 would be too few: each band needs a zip tie loop at both ends)
Rubber bands	**4**	1 per chain, 2 per side, all the same size
Drill, file, smooth tape	-	For holes, chamfers, steps
Calipers, ruler, a 71mm ball	-	Hand checks

4. Print settings

- **Infill: 100%, fully solid**
- Material: PETG or PLA+
- Orientation: on its side
- Supports: none
- Brim: 5mm

5. Step by step

1 Print the V2 rail guard

Print **rail_guard_v2.stl** using the print card below. **100% infill, fully solid.**

2 Fix the slot holes first

V2 slot holes are on a **7.0mm** pitch; the plate is on **8mm**. Either print an updated V2 with 8mm pitch, or drill the plate to match V2. See Warning 1.

3 Cut the flex plate

Print the 1:1 template (page scale 100%, check the scale bar). Cut **11 x 13 holes** from the Medium Gridplate with scissors. On the first piece, cut **1 to 2 holes long**, then trim.

4 Bend the plate

Bend 1: 19 deg down, 5 holes (40mm) from the tongue end. **Bend 2:** 19 deg back to flat, 1 hole from the tail end. Bend along the webs between hole rows. A heat gun helps. Expect springback, so bend slightly past 19 deg.

5 Fit the tongue into V2

Slide the 40mm tongue fully into the 1.7mm slot. Do not zip tie it directly; it is held by the elastic chains in Step 6.

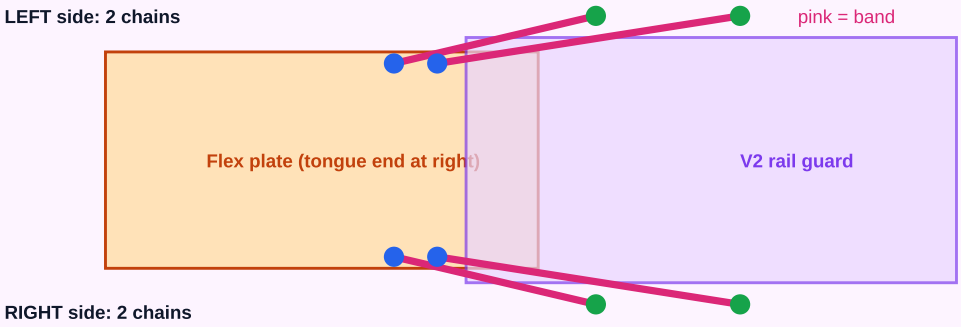
6 Attach with elastic chains (both sides)

On **each side** (left and right) make **2 chains**, 4 in total: **plate hole → zip tie loop → rubber band → zip tie loop → V2 rail guard hole**. The bands pull the tongue fully into the V2 slot and still let the plate flex.

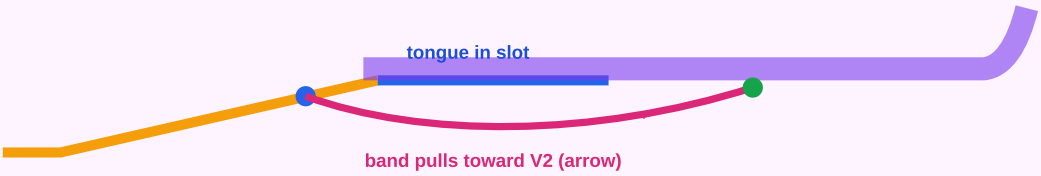
Elastic chain, top view (one chain)



Top view, whole plate: 2 chains per side



Side view (band pulls tongue into slot)



Tips

- **Do not over-tighten** the zip ties. Leave a **loose loop** so the band can stretch.
- Band tension should hold the tongue **fully seated** in the slot.
- Check that **all 4 bands match**. Replace any band that is cracked.
- V2 slot pitch is 7mm and the plate is 8mm, so **pick holes that line up**.
- Trim zip tie tails so they do not touch the ball.

7 Mount V2 and rest the tail

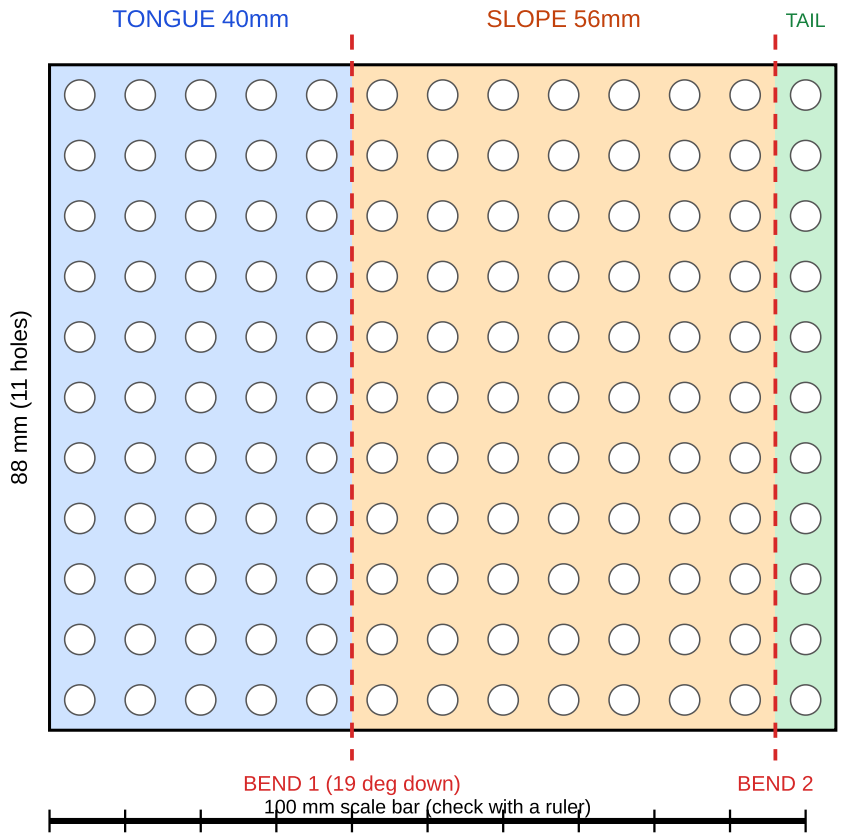
Mount V2 in place on the shooter. The tail rests on the floor plate, ending near y -98.4 (floor edge y -100).

8 Hand check

Run the checklist below before using it on a robot.

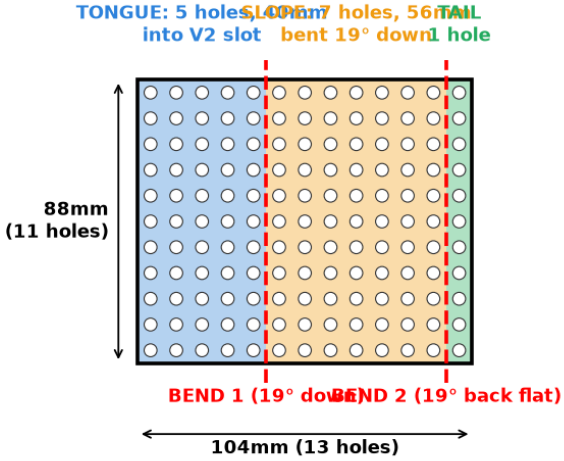
6. Cutting template (1:1)

Print at **100% scale**. Measure the scale bar: it must be 100mm. Blue = tongue, orange = slope, green = tail, red dashed = bend lines.



Flat plate diagram

Flex plate: 11 x 13 cut from Medium Gridplate 1117-0088-0352 (1.5mm polycarbonate, 8mm grid)



7. Known issues (read before building)

1. **Hole pitch mismatch**
   V2 slot holes are 7.0mm apart (x +-40, y -35.5, -29.1, -21.9, -14.6, -7.6, -0.6). The plate is 8mm. Only 2 to 3 line up. Update V2 to 8mm pitch or drill the plate. For the chains, pick holes that line up.

2. **1.1mm step at V2 lip**
   From the 1.03mm top skin over the tongue. Chamfer the edge or cover it with smooth tape.

3. **1.5mm step onto tail**
   The ball steps up from the floor onto the plate tail. Chamfer or tape the tail edge.

4. **Ball coasts undriven, uphill**
   About 131mm with nothing driving it. It needs about 0.7 m/s leaving the intake roller, or a push from the next ball.

5. **Possible clash with kit part 368**
   An upright plate at y -62 may cross the ball path in CAD. Check on the real robot.

8. Hand-check checklist

- [ ] V2 overall size within **+-0.5mm** of 129.35 x 148.57 x 88mm
- [ ] Slot accepts the 1.5mm plate, tongue goes fully 40mm in
- [ ] Bends hold near 19 deg after springback
- [ ] Roll the 71mm ball by hand along the whole path: no catch at the tail or lip
- [ ] All 4 chains fitted, 2 per side, bands match
- [ ] Pull the plate gently away from V2: it should **spring back** fully into the slot
- [ ] No clash with kit part 368 near y -62

9. V2 + flex plate vs V5b

	V2 + flex plate	V5b
Size	129.35 x 148.57 x 88mm	249.4 x 99.7 x 88mm
Material	55 cm3	227 cm3
Squeeze	14.5%	12.6%
Parts	2 (print + cut, bent plate)	**1 print**
Team-to-team variation	Higher (hand bends spring back)	**Low**

Honest note: for untested manufacture for many teams, we think V5b is the lower-variance choice. V2 + flex plate uses less material and is easy to adjust.

Prepared by Aarav J. Feedback from teams is very welcome.